

Load Interconnection System Impact Study Report

For Queue Project Q11-0087

Walney 138 kV

July 02, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0087. The Interconnection Customer proposes a 109.9 MW large load (data center) facility interconnecting at the Walney 138 kV substation in the Virginia area. The request was received on December 13, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 71 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$18,956,000.

3. Network Impacts

3.1 Load Deliverability Analysis

1. Facility: Castleton Energy Center to Dranesville Substation 138kV Circuit 2
Contingency: Castleton Energy Ctr to Dranesville 138kV Line 1
Loading: 135.7% (Rating: 286 MVA)
DFAX: 48.1%
Pre-project loading: 117.2%
2. Facility: Castleton Energy Ctr to Sully 138kV L
Contingency: Fountain to Pleasant Vly 138kV L
Loading: 133.2% (Rating: 232 MVA)
DFAX: 73.8%
Pre-project loading: 98.2%
3. Facility: Castleton Energy Center to Dranesville 138kV
Contingency: Dranesville to Castleton Energy Ctr 138kV Cir. 2
Loading: 127.3% (Rating: 305 MVA)
DFAX: 48.1%
Pre-project loading: 110.0%
4. Facility: Waney to Sully 138kV
Contingency: Fountain to Pleasant Vly 138kV
Loading: 127.1% (Rating: 226 MVA)
DFAX: 73.8%
Pre-project loading: 91.1%
5. Facility: Pleasant Vly to Fountain 138kV Line
Contingency: Castleton Energy Center to Sully Substation 138kV
Loading: 113.0% (Rating: 266 MVA)
DFAX: 57.1%
Pre-project loading: 89.4%
6. Facility: Grand Blanc Generating Station to Cub-Run Dp 138kV
Contingency: Castleton Energy Ctr to Sully 138kV
Loading: 105.6% (Rating: 239 MVA)

DFAX: 57.1%

Pre-project loading: 79.4%

7. Facility: Cub-Run Dp to Pleasant Vly 138kV Line

Contingency: Castleton Energy Ctr to Sully 138kV Line

Loading: 101.2% (Rating: 278 MVA)

DFAX: 57.1%

Pre-project loading: 78.7%

8. Facility: N Anna to Louisa Generation Facility 345kV

Contingency: Louisa Generation Facility to Carrabelle 345kV

Loading: 100.6% (Rating: 1,686 MVA)

DFAX: 5.8%

Pre-project loading: 100.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Dranesville to Castleton Energy Center 138kV Line 2	Rebuild line with higher-capacity conductor	\$7,217,000	88% (\$6,365,000)
Castleton Energy Ctr to Sully 138kV L	Rebuild line with higher-capacity conductor	\$1,707,000	100% (\$1,707,000)
Dranesville to Castleton Energy Ctr 138kV Cir. 1	Rebuild line with higher-capacity conductor	\$2,557,000	88% (\$2,255,000)
Walney to Sully 138kV L	Rebuild line with higher-capacity conductor	\$2,818,000	100% (\$2,818,000)
Pleasant Vly to Fountain 138kV Line	Rebuild line with higher-capacity conductor	\$3,825,000	91% (\$3,489,000)
Grand Blanc Generating Station to Cub-Run Dp 138kV	Reconductor line	\$996,000	91% (\$909,000)
Cub-Run Dp to Pleasant Vly 138kV Line	Reconductor line	\$987,000	91% (\$901,000)
N Anna to Louisa Generation Facility 345kV	Reconductor line	\$33,343,000	2% (\$512,000)
	Total allocated to project		\$18,956,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.